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Blessing or peril? The impact of Artificial Intelligence on the energy transition—evidence from China

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ABSTRACT

The energy transition is essential for combating climate change and achieving sustainability. Artificial intelligence, as a 'double-edged sword' technology, not only provides new momentum for the energy transition but also poses risks of uneven development. This paper investigates the impact of AI on the energy transition using data from 261 cities in China. The findings indicate AI can reduce energy consumption, enhance energy efficiency, and optimize the energy structure, thereby promoting the energy transition across three dimensions: quantity, quality, and structure. The mechanism analysis confirms that AI can indirectly promote the energy transition by improving human-machine alignment and fostering economic agglomeration. Moderation results indicate that the quality of digital infrastructure, technological absorptive capacity, and production factor structure significantly influence the effectiveness of AI. Threshold effects reveal that the impact of AI on the energy transition exhibits an increasing marginal effect under different levels of energy equity. Spatial spillover analysis reveals a significant spatial dependence in the energy transition. However, due to the predominance of the siphoning effect over the trickle-down effect, the impact of AI on neighbouring areas temporarily exhibits a dampening effect. This study enhances the understanding of opportunities presented by AI, providing important references for promoting energy transition and ecological civilization construction.

KEYWORDS

Artificial intelligence; energy transition; spatial spillover; China

JEL CLASSIFICATION

Q42; Q55

I. Introduction

The energy transition is fundamental for the shift from an industrial to an ecological civilization and represents a strategic choice for sustainable global development. In recent years, the environmental crises caused by large-scale fossil energy consumption have become increasingly evident (Bidwell and Sovacool 2023), posing major constraints on socio-economic sustainability. The energy transition is thus essential not only for steady growth but also as a necessary response to the climate crisis. Controlling total consumption, improving efficiency, and optimizing the energy mix to build a clean, secure, and efficient energy system have become a global consensus. As the world's largest developing country and energy consumer, China has actively responded to shifting global energy dynamics. With a focus on energy security, the country has set binding targets to reduce fossil fuel dependence and lower energy intensity. Despite resource constraints characterized by being 'rich in coal but poor in oil and gas', China has reduced its coal consumption share from 72.76% in 2005 to 55.3% in 2023, expanded its renewable energy capacity to over 1.45 billion kilowatts, and decreased its energy intensity to 0.48. Tensions between the path dependence of the energy system and the 'creative destruction' brought by technological diffusion persist, resulting in challenges such as inefficiency, imbalance, and structural rigidity (Khan and Su 2023). In this context, achieving a disruptive transformation of the energy system has become a critical global task for sustainable economic and social development.

Artificial intelligence (AI) is a key driver of energy development and a catalyst for the new industrial revolution, introducing innovative technologies, models, and paradigms for ecological construction (Soori, Arezoo, and Dastres 2023). Defined as machines capable of perception,

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reasoning, and learning (M. Lee 2020), AI is vital in transforming the energy sector and enhancing national energy competitiveness (Tao et al. 2018). Compared with traditional technologies, AI integrates supply and demand, optimizes operations, and lowers costs, thereby promoting the energy transition (B. Wang et al. 2024). However, its reliance on high-intensity computing may also increase energy consumption and emissions. Training large AI models can emit as much carbon as five cars do over their entire lifespans (Strubell, Ganesh, and McCallum 2020). Excessive electricity consumption in data centres may lead to rebound effects, thereby undermining energy savings. Thus, AI is a double-edged sword: improper use could trigger 'green swan' events and systemic risks (Campiglio et al. 2023). Assessing the role of AI in the energy transition is essential for tackling climate change and advancing green development.

The academic debate over whether AI is a boon or a risk to the energy transition is intensifying, yet it remains at an impasse, as reflected in three key issues. First, there is analytical fragmentation: most studies focus solely on AI's impact on the energy efficiency (P. Zhao, Gao, and Sun 2022), while overlooking its potential in consumption control and structural optimization. Second, mechanism opacity: existing research rarely uncovers how AI affects the energy transition, especially under varying external conditions. Third, spatial neglect: conventional studies assume spatial independence and overlook the regional spillover effects of AI.

Building on these gaps, this study poses the question: Does AI drive China's energy transition through the three-dimensional pathway of quantity suppression, efficiency improvement, and structural optimization? What are the underlying mechanisms of its influence? Does AI exhibit spatial spillover effects? To answer these questions, the study uses panel data from 261 Chinese cities to examine AI's impact and mechanisms on the energy transition, explores its heterogeneous effects, and incorporates spatial analysis to assess potential spillovers.

The key contributions of this paper include: First, this study dissects the core dimensions of the energy transition through a threefold lens: quantity, efficiency, and structure. Furthermore, it integrates AI with the energy transition, aligning the threefold dimensions of the energy transition with the core functional drivers of AI. Within this integrated framework, the influence of AI on the energy transition is examined, thereby enriching and extending the literature on AI's environmental benefits in the energy sector. Second, it reveals how AI shapes the mechanisms of the energy transition and investigates its moderating role from a novel perspective. Specifically, it identifies for the first time the moderating effects of digital infrastructure, technological absorptive capacity, and the structure of production factors. Finally, the study investigates the spatial spillover effects of AI on energy transition, further enriching the relevant research. While existing literature mainly focuses on the temporal and panel effects of AI, it often overlooks its characteristics of permeability, integration, and synergy (C.-C. Lee et al. 2024; T. Li et al. 2024). This study deepens the understanding of regional interactions and the diffusion of AI across contexts, offering new insights for the policy discourse on AI and green energy.

II. Literature review

The fundamental driver of ET lies not only in productivity gains but also in resolving tensions between economic growth and escalating environmental, climate, and security challenges (Ullah et al. 2023). ET refers to replacing traditional energy sources with cleaner alternatives to enhance efficiency and promote sustainability (Chishti et al. 2024). Existing studies identify three main drivers of low-carbon energy transitions. First, policy-driven factors. Hassan et al. (2024) found that ecological and technological assistance policies promote the adoption of clean energy. Second, finance-driven factors, where financial investments are pivotal in the progress of ET (Nepal et al. 2024). Z. Wang et al. (2024) found that digital finance boosts the efficiency of ET and exerts a spatial effect. Third, technology-driven factors. Innovation drives clean and low-carbon transformation of energy systems (Dogan et al. 2022; Luo et al. 2024). Chishti and Dogan (2024) argue that ICT technology is a key enabler of renewable energy.

Since the emergence of AI, there has been extensive scholarly discussion on its economic and social

impacts. Researchers have explored various aspects of AI, including its effects on supply chains (Modgil et al. 2021), economic cycles (Chishti, Dogan, and Binsaeed 2024), and production efficiency, generally concluding that AI reshapes industry structures, boosts economic efficiency, and improves development prospects. For instance, Xia, Han, and Yu (2024) found that industrial intelligence promotes both the scale and quality of industrial upgrading. Other studies focus on AI's effects on the labour market, arguing that it accelerates automation, displaces labour in the short term (Aghion, Jones, and Jones 2017), and increases labour productivity (Graetz and Michaels 2018).

Recently, scholars have increasingly incorporated AI and the energy environment into a unified analytical framework. Some researchers argue that AI is vital in the energy sector. Using data from 67 countries, Q. Wang, Li, and Li (2024, Q. Wang, Zhang, Li, Sun 2024) found that AI facilitates ET, with threshold effects under varying trade openness. Q. Wang, Li, and Li (2024), using global data, revealed that AI's effect on ET is uncertain in the short term, suppressive in the medium term, and positive in the long term. Q. Wang, Zhang, and Li (2024) showed AI has the greatest impact on sustainable development in mid-to-high sustainability countries. However, some scholars highlight AI's dual role. While improving efficiency, AI also drives energy consumption through data centre scale effects (Zou, Yang, and Feng 2023). The crowding-out effect of digital industry development on CO_2 offsets efficiency gains. For example, Y. Zhang et al. (2025), using data from G20 countries, found that AI exacerbates energy vulnerability in less-developed economies.

The existing literature has several shortcomings: First, spatial mismatches in research scales. There is no consensus on whether AI promotes or hinders the energy transition, most studies focus on the national or provincial level. In contrast, this study employs city-level data to conduct a more granular analysis. Second, fragmentation in indicator systems. Existing research often examines AI's impact through single-dimensional indicators, which risks isolating its effects on energy structure optimization and efficiency improvement, while ignoring potential synergies or trade-offs. Third, the black-box nature of impact mechanisms. Limited identification of the channels and heterogeneous effects through which AI influences energy transition restricts comprehensive understanding and policy design. Finally, the lack of spatial linkage analysis. Although AI can overcome geographic and informational barriers between cities, most studies rely on panel regressions and fail to capture its cross-regional spillover effects.

III. Theoretical basis

The impact of AI on the energy transition

AI has reshaped production modes in the socio-economic landscape, enhancing economic quality and promoting a low-carbon energy transition. One primary approach to achieving energy transition is reducing energy consumption. By utilizing advanced information technology systems, AI can track energy transactions and consumption instantly. This enables the rational allocation of energy resources, which in turn helps reduce overall energy consumption (Abdalla et al. 2021). Additionally, AI can optimize energy systems and establish intelligent microgrids, thus improving energy management. AI enables governments to efficiently collect and process energy data, enabling more scientific decisions (M. Zhang and Liu 2022). Furthermore, companies can gain insights into market trends and maximize resource utilization to reduce energy consumption. Finally, AI can promote energy-efficient products and practices, reshaping consumer awareness and reducing energy consumption (Nudurupati et al. 2022).

Improving energy efficiency is a crucial pathway to achieving the energy transition. AI can generate scale compensation effects, facilitating the transition from traditional to low-carbon technologies and thereby enhancing energy efficiency (Koch, Manuylov, and Smolka 2021). As AI integrates into environmental protection, it drives digital advancement in the energy market. AI-based monitoring of precise energy measurements enhances transparency and accuracy in energy usage, leading to better resource management and increased energy efficiency (Y. Liu, Zhang, and Shen 2024). Additionally, integrating industrial robots into

production processes optimizes resource allocation and improves energy efficiency.

Optimizing the energy structure is a primary pathway to achieving the energy transition. AI can enhance the generation efficiency and distribution of renewable energy, thereby increasing the share of clean energy (Pradhan and Ghosh 2022). Additionally, AI enables investors to more accurately assess the potential returns and risks of projects, attracting investment in clean energy. AI can analyse vast amounts of data to guide capital flows towards environmentally friendly enterprises, thereby facilitating ET (Sinha et al. 2023). Moreover, AI facilitates the incorporation of intelligent components into business operations, enhancing the shift from tangible assets to digital alternatives and lessening dependence on high-emission energy sources such as coal. The framework diagram illustrating these mechanisms of action is shown in Figure 1. Therefore, we propose:

Hypothesis 1: AI can reduce energy consumption, improve energy efficiency, and optimize the energy structure, thereby promoting the energy transition through a three-dimensional framework of 'quantity – quality – structure'.

Mechanisms of AI's impact on the energy transition

Human-machine alignment is a key factor in driving energy transition. AI can replace labour in performing routine tasks, thereby automating daily operations. Additionally, AI enhances productivity by leveraging skilled labour, reducing energy consumption without compromising output (Aghion, Jones, and Jones 2017). AI evolves from manual to automated operations and from basic analysis to processing complex data, thereby enabling intelligent decision-making (E.-Z. Wang, Lee, and Li 2022). In this new division of labour, workers shift from traditional machine operators to more cognitive roles, intelligent machines undertake autonomous learning and processing, forming collaborative units with workers for complex and high-risk tasks. This model optimizes the complementary functions between AI and workers (Acemoglu and Restrepo 2022; Acemoglu et al. 2022), leveraging their strengths to improve productivity. Consequently, we propose:

Hypothesis 2a: AI can optimize the complementarity between AI and labour by enhancing the matching efficiency with the workforce, thereby indirectly facilitating the energy transition.

Economic agglomeration is a key driver of the energy transition. With the maturation of AI technology, its timeliness and flexibility help address precise allocation issues across the supply chains (Lin and Xu 2024). AI enables data transmission and information dissemination, attracting highly skilled labour and industries. The integration of skilled labour and AI induces the clustering of high-end talent, creating a positive cycle that fosters economic agglomeration. Through technological spillovers, economic agglomeration drives innovation, optimizes resource allocation, and enhances energy efficiency (Qin et al. 2023). Additionally, the synergy generated by economic agglomeration produces knowledge spillovers, reduces transaction costs, fosters an open environment, and promotes the energy transition (K. Liu et al. 2022). Therefore, we propose:

Hypothesis 2b: AI can attract high-skilled labour and industrial inflows, foster economic agglomeration, and indirectly drive the energy transition.

The spatial spillover effects of artificial intelligence on the energy transition

AI can influence the energy transition of neighbouring areas through siphoning effects. AI core areas with favourable conditions attract resources and form growth poles (Barata 2019). This concentration of resources in AI-advantaged areas may decrease overall resource allocation efficiency. Consequently, surrounding cities with fewer growth drivers and incentives may face a widening gap in AI development, leading to a siphoning effect that hinders their energy transition (X. Zhao et al. 2022).

On the other hand, AI may promote the energy transition in neighbouring areas through the

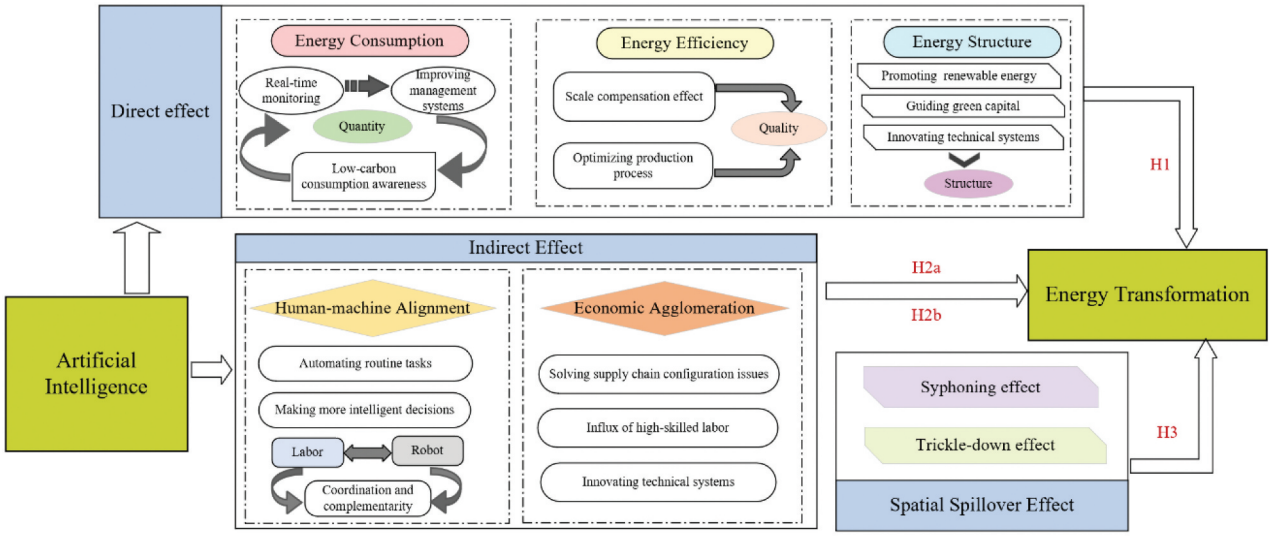


Figure 1. Mechanism of action Diagram.

trickle-down effect. Resource surpluses from upgraded core areas diffuse to peripheral ones (Acs et al. 2021). These peripheral regions benefit from technological spillovers via shared networks and platform economies, which promote technological and industrial upgrading, enhance energy efficiency, and facilitate the energy transition. The impact of AI on the energy transition in neighbouring regions depends on the relative strength of siphoning and trickle-down effects. Existing studies largely suggest that the clustering effects induced by AI tend to strengthen the siphoning effect (Guo, Peng, and Luo 2025; J. Xu and Yeh 2005). Therefore, we propose:

Hypothesis 3: AI exhibits spatial spillover effects on the energy transition. Due to stronger siphoning effects, AI is likely to suppress the energy transition in neighbouring regions.

IV. Model construction and variable selection

Variable selection

Explained variables

Energy transition represents a significant global challenge due to its multifaceted complexity. Current research often examines energy transition from a single dimension. However, to evaluate and advance energy transition comprehensively, this paper explores three key aspects: energy consumption (*EC*), energy efficiency (*EE*), and energy structure (*ES*). First, *EC* is a crucial indicator of regional energy demand, reflecting the scale and growth trends of energy needs. This study utilizes the methodology of Lin and Huang (2023) and calculates *EC* by converting city-level consumption of electricity, liquefied petroleum gas, and natural gas. Second, *EE* reflects technological progress and the rationality of energy resource utilization, measured in this study through the GDP to energy consumption ratio, based on the method of J. Li et al. (2023). Lastly, *ES*, vital for a low-carbon transition, involves the proportion and shifts among different energy types within the energy mix, measured here by the ratio of coal consumption to clean energy consumption. To mitigate the influence of extreme values, the variables are winsorized at the 1st and 99th percentiles.

Figures 2 to 4 show that from 2006 to 2019, China's EC, EE, and ES improved significantly in the central and eastern regions, driven by a strong industrial base, high-tech industries, and rising energy demands. Economic growth and technological innovation boosted green technology adoption, enhancing EE and ES. In contrast, the

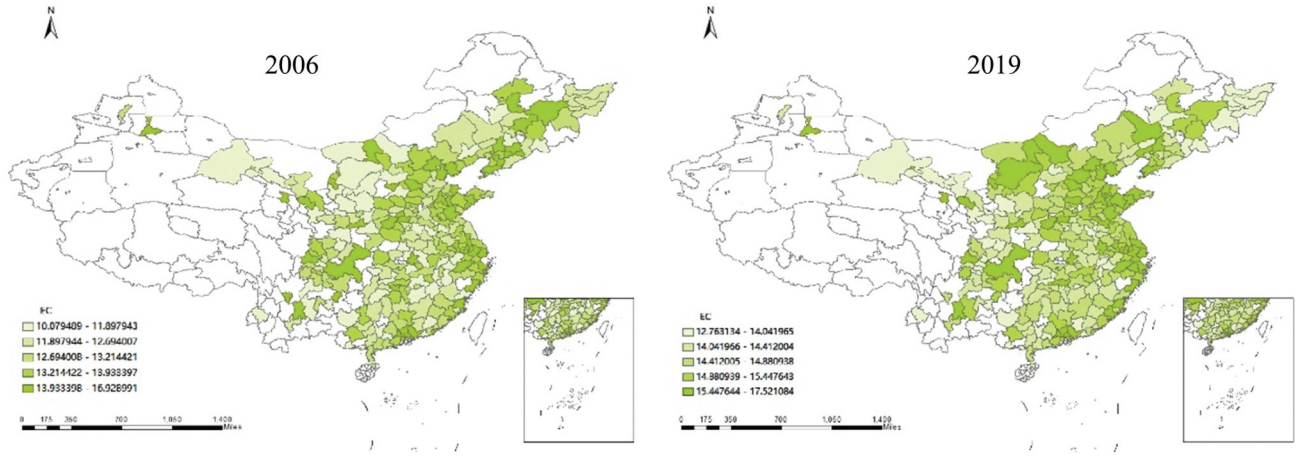


Figure 2. Energy structure (ES) across different cities in 2006 (left) and 2019 (right).

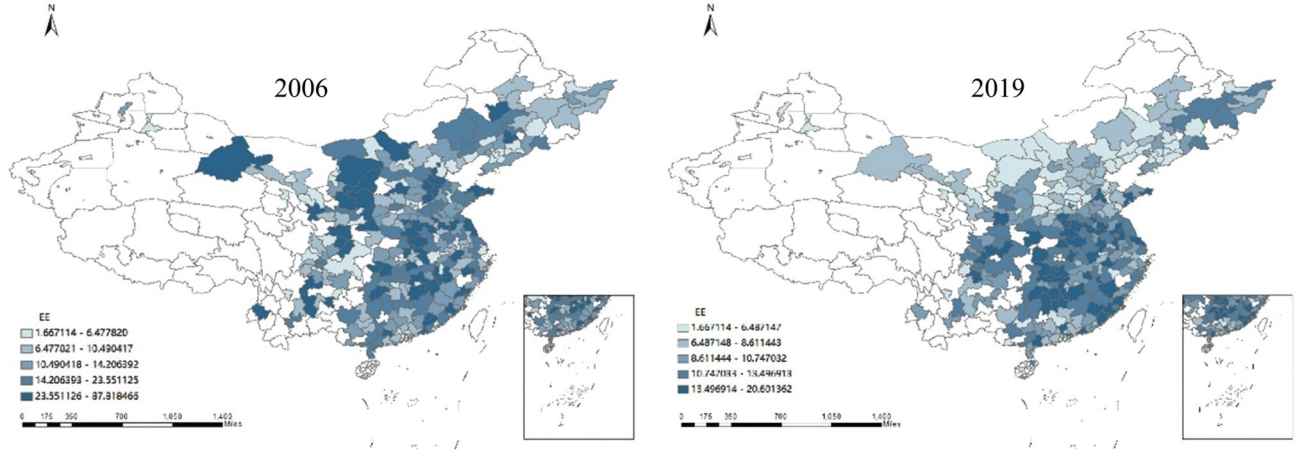


Figure 3. Energy efficiency (EE) across different cities in 2006 (left) and 2019 (right).

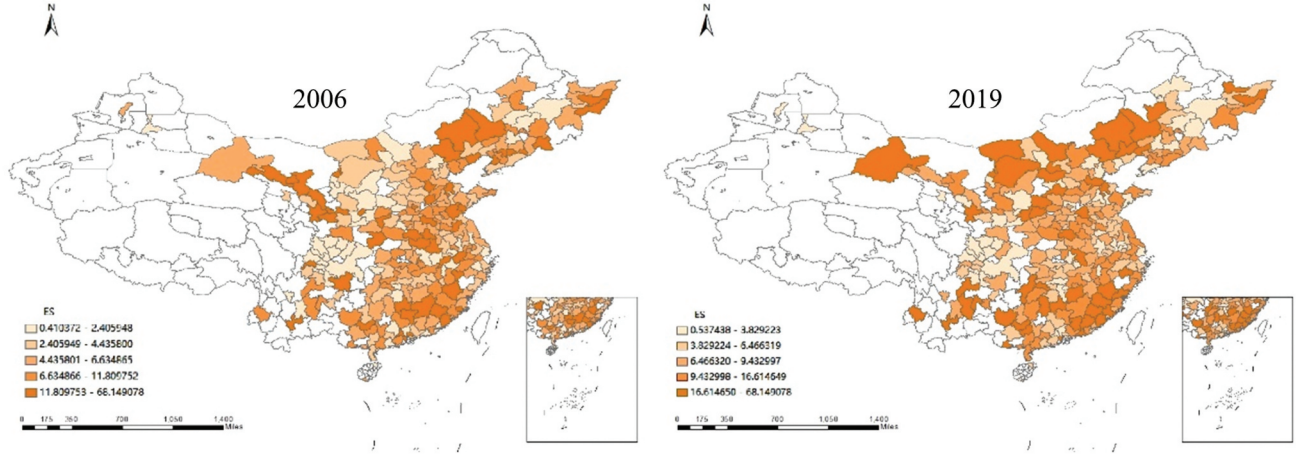


Figure 4. Energy structure (ES) across different cities in 2006 (left) and 2019 (right).

western regions lagged due to weaker industrialization, reliance on polluting industries, and inadequate environmental policies.

Core explanatory variable

Artificial Intelligence (*AI*). Quantifying AI requires overcoming the challenge of technological unobservability. This study uses robot density as a proxy, based on two key considerations: observability and technological generational change. From the observability perspective, industrial robots represent a tangible and measurable manifestation of AI in the real economy. The IFR dataset provides detailed micro-level data on AI applications across sub-industries in China. From the perspective of technological transformation, AI-enabled robots operate adaptively in dynamic environments. Rising robot density reflects not only greater automation but also a shift towards data-driven, autonomous decision-making systems, inherently tied to AI's broader influence beyond traditional automation. Following the methods of Acemoglu and Restrepo (2020) and J. Xu et al. (2022), the study uses the following formula to measure AI:

$$AI_{it} = \sum_{s=1}^{S} \frac{employ_{s,i,t=2008}}{employ_{i,t=2008}} \times \frac{Robot_{st}}{employ_{s,t=2008}} \quad (1)$$

where, s represents the collection of various manufacturing industries, i and t respectively represent city and year. *Robot* is the stock of robots. *employ* is the number of employees. $employ_{s,i,t=2008}/employ_{i,t=2008}$ represents the ratio of the employment in industry s to the total employment in city i, serving as the weight for the robot density in each industry. $Robot_{st}/employ_{s,t=2008}$ is the robot density indicator at the industry level for each year.

As shown in Figure 5, the distribution maps of AI in China from 2006 to 2019 indicate overall improvement across regions. The eastern coastal areas remain leaders, driven by strong economies, advanced technology, and robust policy support. The central and northern regions have significantly reduced regional disparities, while the western areas continue to lag due to development imbalances. To ensure more balanced growth, enhanced policy support and resource allocation are necessary.

Mediating variables

Human-Machine Alignment (*HMA*): *HMA* represents the degree of matching between AI and labour capability. AI can enhance the close cooperation between workers and machines, achieving complementary synergy that promotes ET. This paper uses a coordinated coupling model to calculate the degree of human-machine matching (Hu, Wang, and Zhao 2021).

$$HMA = \sqrt{U \times V} \quad (2)$$

Where, $U = 2\sqrt{AI \times hum}/(AI + hum)$, $V = 0.5 \times AI + 0.5 \times hum$. *hum* is represented by the proportion of employees in the information transmission, computer services, and software industries. Both *AI* and *hum* have been standardized using the range standardization method.

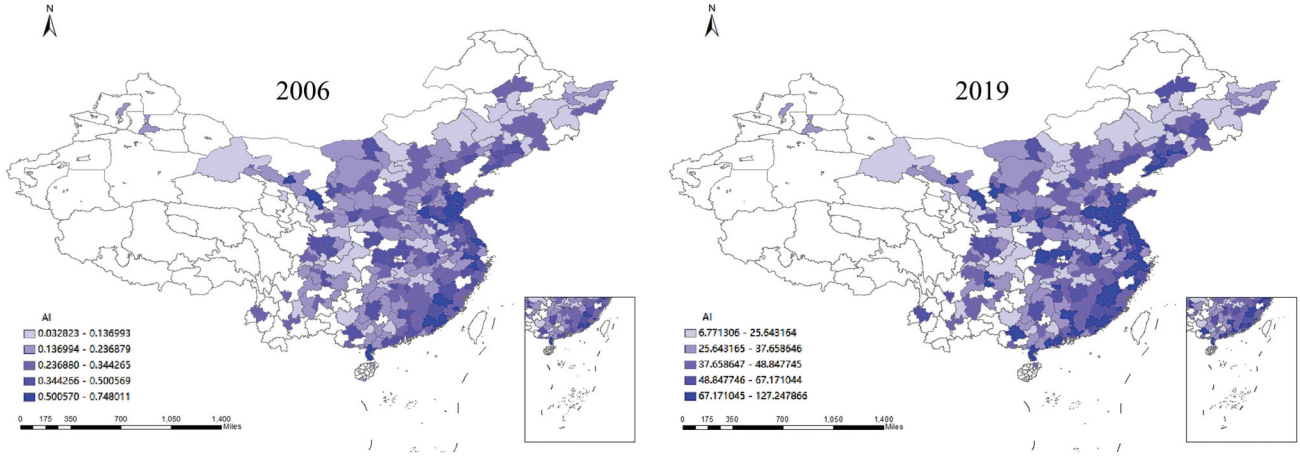


Figure 5. Artificial Intelligence(AI) across different cities in 2006 (left) and 2019 (right).

Economic Agglomeration (*EA*): Scholars often measure economic agglomeration using economic density (M. Xu, Tan, and He 2022). *EA* is measured by the ratio of the added value of non-agricultural industries to the administrative area.

Moderating variables

Digital Infrastructure (*infra*): The functionality of AI requires support from digital information infrastructure. Digital Infrastructure is quantified by the proportion of internet broadband access users to the population (L. Wang and Shao 2024).

Technological Absorptive Capacity (*TAC*): TAC is essential for technological catch-up and leveraging the latecomer advantage (Yin and Zeng 2023). This study measures *TAC* by the ratio of scientific expenditure to fiscal expenditure.

Production Factor Structure (*PFS*): The impact of AI on ET varies under different *PFS*. This study measures *PFS* by the ratio of labour to capital stock (B. Wang et al. 2021).

Threshold variable

Energy Equality (*EEQ*). Following the approach of S. Zhang et al. (2021), this study adopts the economic contribution coefficient to evaluate the energy equality of each city from an economic perspective. If a region's share of energy consumption exceeds its share of GDP, it is considered to infringe upon the interests of other regions. Therefore, *EEQ* measures regional energy equality by evaluating the balance between energy consumption and economic output. The formula is as follows:

$$EEQ = \frac{GDP_i/GDP_t}{EC_i/EC_t} \tag{3}$$

Where, GDP_i and GDP_t represent the GDP of city i and the national GDP, respectively. EC_i and EC_t denote the energy consumption of city i and the national energy consumption, respectively. A higher *EEQ* index indicates a higher level of energy equality.

Control variables

Given the numerous variables that influence energy transition, the following control variables are selected based on relevant literature (Nepal et al. 2024): Economic Development Level (*lngdp*): The logarithm of per capita GDP. Government Size (*gov*): Local fiscal expenditure/GDP. Industrial Structure (*is*): 1× primary industry proportion + 2× secondary industry proportion + 3× tertiary industry proportion. Transportation Infrastructure (*lntra*): The logarithm of the number of buses per capita. Financial Development (*fin*): Loan balance/GDP.

Model construction

Baseline model

There is significant cross-sectional and temporal variation across China's cities. To address unobserved heterogeneity and support causal inference, this study uses a two-way fixed effects model that controls for both city effects and year effects to assess AI's impact on the energy transition. The model is specified as follows:

$$Y_{it} = a_0 + a_1 AI_{it} + a_2 X_{it} + \mu_i + \tau_t + \varepsilon_{it} \tag{4}$$

where i and t represent cities and years, respectively. Y represents energy transition, which includes energy consumption (*EC*), energy efficiency (*EE*), and energy structure (*ES*). *AI* represents artificial intelligence. X is the set of control variables. μ_i is the area fixed effect, τ_t is the time fixed effect, and ε_{it} is the random perturbation term.

Mediation model

To further test the hypotheses and examine whether AI influences the energy transition through human – machine matching and economic agglomeration, this study adopts a mediation effect model. Considering the limitations of the traditional three-step approach, the study follows the four-step mediation model proposed by Zeng, Su, and Peng (2023), which adds a regression of the mediator on the dependent variable to improve the completeness of the causal pathway.

$$M_{it} = b_0 + b_1 AI_{it} + b_2 X_{it} + \mu_i + \tau_t + \varepsilon_{it} \tag{5}$$

$$Y_{it} = c_0 + c_1 M_{it} + c_2 X_{it} + \mu_i + \tau_t + \varepsilon_{it} \tag{6}$$

$$Y_{it} = e_0 + e_1 M_{it} + e_2 AI_{it} + e_3 X_{it} + \mu_i + \tau_t + \varepsilon_{it} \quad (7)$$

Where M is the mediator variable, regression on Eq. (5), if b_1 is significantly positive, it indicates that AI promotes M, after adding the mediator variable, if c_1 in Eq. (6) and e_1 in Eq. (7) are significant, it means that the mediation effect are established.

Moderating effect model

Regional external environments may affect the strength of AI's impact on the energy transition. By incorporating an interaction term between AI and the moderating variable, the moderation model identifies how AI's impact varies under different conditions and captures its interaction with environmental factors. Accordingly, this study adopts a moderation model to explore the heterogeneity in AI's effect on the energy transition.

$$Y_{it} = d_0 + d_1 AI_{it} \times N_{it} + d_2 AI_{it} + d_3 N_{it} + d_4 X_{it} + \mu_i + \tau_t + \varepsilon_{it} \quad (8)$$

Where N represents the moderating variables, including *infra*, *TAC*, and *PFS*. If the coefficient d_1 is significant in the regression of Eq. (8), it indicates that the moderating effect is valid.

Threshold model

AI may have a nonlinear impact on the energy transition. To examine this possibility, this study adopts the threshold model developed by Hansen (1999).

$$Y_{it} = \eta_0 + \eta_1 AI_{it}(thr \le v_1) + \eta_2 AI_{it}(v_1 < thr \le v_2) + \cdots + \eta_n AI_{it}(thr > v_n) + \gamma X_{it} + \varepsilon_{it} \quad (9)$$

Where, *thr* is the threshold variable, and $v_1, v_2, \ldots v_n$ is the threshold value to be estimated. The remaining variables are consistent with those defined in the previous sections.

Data sources

Robot data are sourced from the IFR Robot Application Statistics. Data on energy transition and control variables are obtained from the China City Statistical Yearbook, the EPS database, and prefecture-level statistical yearbooks. Based on data availability, panel data from 261 Chinese cities spanning 2006 to 2019 are selected as the empirical sample. Variable definitions and descriptive statistics are presented in Table 1.

V. Empirical results and analysis

Basic regression

Based on the benchmark model, this study analyzes the impact of AI on the energy transition (ET) from the dimensions of quantity, quality, and structure. Columns (1) and (2) of Table 2 show that AI significantly reduces energy consumption, aligning with theories suggesting that technological progress improves efficiency and reduces waste (P. Zhao, Gao, and Sun 2022). As the world's largest energy consumer, China benefits from AI-enabled forecasting and optimization, which help address the 'growth – energy control' dilemma. Columns (3)

Table 1. Descriptive statistics.

Variable	Definition	Obs	Mean	Std. Dev.	Min	Max
EC	Energy consumption	3654	13.716	1.215	9.354	17.521
EE	Energy efficiency	3654	18.117	14.45	1.667	87.818
ES	Energy structure	3654	8.664	10.722	0.410	68.149
AI	Artificial intelligence	3654	13.191	18.556	0.033	127.248
lngdp	Economic development	3654	10.448	0.713	4.595	13.056
gov	Scale of government	3654	0.171	0.085	0.043	1.485
is	Industrial structure	3654	2.269	0.142	1.890	2.832
lntra	Transportation facilities	3654	0.658	1.03	−2.428	4.705
fin	Financial development	3654	0.90	0.562	0.075	7.450
HMA	Human-Machine Alignment	3654	0.249	0.114	0	0.774
EA	Economic agglomeration	3654	0.257	0.654	0.001	13.472
infra	Information infrastructure	3654	17.914	18.54	0.403	366.348
PFS	Production factor structure	3654	0.010	0.006	0.002	0.065
TAC	Technology absorptive capacity	3654	0.015	0.015	0	0.207
EEQ	Energy equality	3654	1.653	1.517	0.024	26.52

Table 2. Baseline regression results.

	(1) EC	(2) EC	(3) EE	(4) EE	(5) ES	(6) ES
AI	−0.0043***	−0.0041***	0.0981***	0.0916***	−0.0832***	−0.0859***
	(0.0009)	(0.0009)	(0.0180)	(0.0173)	(0.0217)	(0.0215)
lngdp		0.3565***		1.0676		0.5755
		(0.0792)		(0.9175)		(0.7471)
gov		0.7768***		−9.5050**		9.7418**
		(0.2200)		(4.2983)		(4.2080)
is		0.7241***		−14.6899***		2.1990
		(0.2027)		(4.8264)		(4.8703)
Intra		0.1485***		−2.1300***		2.1592***
		(0.0319)		(0.6544)		(0.5926)
fin		−0.1081***		1.2462*		−1.8717**
		(0.0384)		(0.6423)		(0.9084)
Constant	13.7730***	8.2700***	16.8222***	40.9857**	9.7616***	−2.6032
	(0.0139)	(1.0026)	(0.2831)	(17.3835)	(0.3115)	(13.8769)
Id FE	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES
N	3654	3654	3654	3654	3654	3654
R^2	0.9205	0.9252	0.7162	0.7197	0.6039	0.6087

Standard errors in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

and (4) reveal a significant positive relationship between AI and energy efficiency, supporting the view that factor reallocation and innovation enhance productivity. AI shifts production towards high-productivity sectors and promotes green manufacturing. Columns (5) and (6) suggest that AI also contributes to optimizing the energy structure, although the effect is weaker than that on efficiency, reflecting China's emphasis on manufacturing process optimization and sectoral variation in AI adoption. Compared with Singapore, a global leader in robot density, both countries demonstrate an AI-driven path towards energy transition (Islam, Shahbaz, and Ahmed 2024). However, differences in industrial structures and energy policies impose distinct institutional constraints on technological pathways. While Singapore emphasizes system-level solutions such as smart grids, China relies on manufacturing-based, localized breakthroughs – offering a scalable model for emerging economies.

Endogeneity test

To avoid bias in the regression estimates, this study employs an instrumental variable approach to test the robustness of the empirical findings. Following Y. Wang and Dong (2020), the penetration index (IV1), constructed using U.S. robotics data, is employed as an instrumental variable. Regarding relevance, the United States maintains a globally leading level of robot application, which reflects the broader trajectory of global technological progress. Its development trend closely parallels that of China during the same period and is strongly correlated with the diffusion of robotics in China. Regarding exogeneity, given the relatively mature energy market in the U.S., sectoral levels of robot application there are unlikely to directly affect China's energy market. Thus, they primarily reflect exogenous technological advancements, satisfying the exclusion restriction. Table 3 shows that the KP-LM statistic and the KP-F statistic confirm the

Table 3. Endogeneity test based on IV1.

	(1) AI	(2) EC	(3) EE	(4) ES
IV1	1.9522***	−0.0038***	0.0689***	−0.0952***
	(0.0266)	(0.0008)	(0.0164)	(0.0199)
Constant	18.5431***	10.2140***	40.8427**	−14.3326
	(6.4045)	(1.0312)	(18.5485)	(15.0061)
Controls	YES	YES	YES	YES
Id FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
N	3654	3654	3654	3654
R^2	0.9780	0.9252	0.7196	0.6087
KP-LM		477.997***		
KP-F		5381.791		

Table 4. Mechanism of action based on HMA.

	(1) *HMA*	(2) *EC*	(3) *EC*	(4) *EE*	(5) *EE*	(6) *ES*	(7) *ES*
AI	0.0012***		−0.0017*		0.0512***		−0.0647***
	(0.0001)		(0.0009)		(0.0172)		(0.0215)
HMA		−2.1541***	−1.9970***	37.6150***	32.8293***	−23.2667***	−17.2241***
		(0.2502)	(0.2600)	(4.4870)	(4.4355)	(3.8932)	(3.5646)
Constant	0.4798***	9.1362***	9.2282***	28.0356	25.2332	2.1230	5.6614
	(0.0633)	(0.9667)	(0.9653)	(17.7542)	(17.6282)	(13.7006)	(13.7238)
Id FE	YES	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES	YES
N	3654	3654	3654	3654	3654	3654	3654
R^2	0.9428	0.9271	0.9272	0.7228	0.7236	0.6086	0.6106
Sobel Z		−0.157**		4.786***		−6.043***	

rationality and feasibility of the instrumental variable selection. The results remain consistent with previous findings.

Additionally, Appendix A employs alternative instrumental variables to further address potential endogeneity concerns. Appendix B tests the robustness of the results using four approaches: replacing the dependent variable, replacing the independent variable, adjusting the sample size, and applying robust standard error adjustments.

Mechanism of action

This study explores whether AI influences the energy transition through human-machine alignment (HMA) and economic agglomeration mechanisms. The mediation analysis for HMA (Table 4) reveals that AI facilitates the energy transition by enhancing HMA. The core of this mechanism lies in the AI-driven transformation of the workforce towards higher-skilled labour and more efficient resource allocation (Binder and Bound 2019). For instance, with the development of AI, frontline workers at Mondelēz's Beijing plant have gradually assumed technician roles, leading to a 10%–25% reduction in energy consumption.[1] Similarly, at CATL, the integration of AI-based visual inspection and human collaboration has significantly reduced both defect rates and energy consumption per production line. Cities with an HMA advantage can maximize the complementary and coordinated use of workers and machines, enhancing resource allocation efficiency and promoting energy transition (Xie et al. 2021). Overall, AI indirectly promotes the energy transition by improving human-machine alignment, thereby supporting hypothesis H2a.

The economic agglomeration (EA) results (Table 5) indicate that AI promotes the energy transition by fostering agglomeration effects. As AI advances, reduced information costs attract firms

Table 5. Mechanism of action based on EA.

	(1) *EA*	(2) *EC*	(3) *EC*	(4) *EE*	(5) *EE*	(6) *ES*	(7) *ES*
AI	0.0032***		−0.0038***		0.0855***		−0.0838***
	(0.0006)		(0.0009)		(0.0173)		(0.0217)
EA		−0.1220***	−0.1120***	2.1159***	1.8897***	−0.8735***	−0.6519**
		(0.0346)	(0.0329)	(0.6230)	(0.5834)	(0.2909)	(0.2641)
Constant	2.4297***	8.1453***	8.5421***	45.3774**	36.3943**	−9.8174	−1.0193
	(0.5379)	(1.0080)	(0.9971)	(17.6965)	(17.4643)	(14.0744)	(13.9609)
Id FE	YES	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES	YES
N	3654	3654	3654	3654	3654	3654	3654
R^2	0.8219	0.9252	0.9258	0.7188	0.7210	0.6052	0.6090
Sobel Z		−0.010***		0.226***		2.644***	

[1]Data Source: McKinsey Global Institute, https://www.mckinsey.com.cn/category/insights/mckinsey-global-institute/.

and talent, further reinforcing agglomeration effects. This boosts innovation, improves allocation, and accelerates the low-carbon transition. AI adoption reshapes industrial layouts and information flows, enabling innovative firms to cluster and form ecosystems. For example, Shenzhen's AI growth has made it an innovation hub, attracting AI firms and talent. This concentration generates agglomeration externalities that enhance energy efficiency (Fang et al. 2020). EA also lowers clean energy prices by promoting industrial upgrading and green innovation, thereby stimulating demand. Germany's 'Industry 4.0' initiative promotes digital and green technologies, supporting renewable energy development and energy mix optimization. Overall, AI promotes energy transition by improving EA, thereby confirming hypothesis H2b.

Moderating effects

The previous sections examined the impact of AI on ET from a full-sample perspective. However, the effect may vary under different policy environments and market attributes. Consequently, this study further investigates the moderating effects of digital infrastructure (*infra*), technological absorptive capacity (*TAC*), and production factor structure (*PFS*).

Digital infrastructure

This study examines how digital infrastructure (Infra) moderates the impact of AI on the ET. Table 6 shows that *Infra×AI* is significantly negative for EC and ES, but positive for EE. This indicates that advanced Infra enhances the effectiveness of AI in promoting ET. In such regions, intelligent production is integrated into industrial processes, thereby reducing energy consumption (Huang, Shen, and Yao 2024). Moreover, digital infrastructure strengthens smart governance and enables full AI utilization. It reduces information asymmetry in government energy management, improving efficiency. For example, the Transmission Facilitation Program under the 2021 U.S. Infrastructure Act optimized wind energy transmission, thereby enhancing energy distribution efficiency. Conversely, underdeveloped infrastructure limits AI deployment due to facility gaps, technical constraints, and high costs, weakening its impact on the energy transition. This supports the view of Hong, Shi, and Zheng (2023), who view digital infrastructure as a key to green development. Therefore, targeted investment in Infra is crucial to enabling intelligent production, improving operating conditions, and fostering an AI-friendly environment for an effective energy transition.

Technological absorptive capacity

This study explores the moderating effect of TAC on the impact of AI on the energy transition. In high-TAC regions, AI significantly reduces EC and enhances EE, but has limited influence on optimizing ES (Table 6, Columns 4–6). Stronger research funding leverage in these areas supports intelligent equipment adoption, streamlines

Table 6. Results of moderating effects.

	(1) EC	(2) EE	(3) ES	(4) EC	(5) EE	(6) ES
Infra×AI	−0.0001***	0.0028***	−0.0010***			
	(0.0000)	(0.0004)	(0.0004)			
TAC×AI				−0.0371*	1.3211***	0.4070
				(0.0225)	(0.4067)	(0.4223)
AI	−0.0004	−0.0147	−0.0471*	−0.0033***	0.0600***	−0.0949***
	(0.0011)	(0.0231)	(0.0255)	(0.0010)	(0.0191)	(0.0282)
infra	0.0008	−0.0281**	−0.0190*			
	(0.0005)	(0.0111)	(0.0111)			
TAC				0.5765	−20.4269	−24.0299*
				(0.7551)	(14.3152)	(13.0885)
Constant	8.8335***	25.5339	4.6188	8.3318***	38.7836**	−3.1153
	(0.9962)	(17.7159)	(14.1258)	(1.0074)	(17.3735)	(13.9672)
Controls	YES	YES	YES	YES	YES	YES
Id FE	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES
N	3654	3654	3654	3654	3654	3654
R^2	0.9258	0.7231	0.6103	0.9253	0.7204	0.6089

production, and boosts energy productivity. For example, the EU's 2023 Green Industrial Plan and Net-Zero Industry Act aim to accelerate the transition by strengthening R&D and clean-tech manufacturing. In contrast, low-TAC regions face shortages in human capital and investment, constraining AI's application in energy transition (Yin and Zeng 2023). ES exhibits strong path dependence shaped by economic and social factors, making structural change difficult despite AI's potential. Overall, the findings suggest that AI is more effective in promoting the energy transition in regions with high TAC.

Production factor structure

This study investigates whether the impact of AI on ET varies with labour intensity. Table 7 shows that the interaction terms PFS×AI are significantly negative for EC and ES, and significantly positive for EE, suggesting that AI is more effective in promoting ET in labour-intensive regions. In labour-intensive areas, AI's substitution effect on labour is more evident (Fan, Hu, and Tang 2021). Robotic operations enhance precision, reduce manual inefficiencies, and improve energy efficiency. In contrast, capital-intensive regions dominated by energy-intensive industries may face higher energy demand due to the electricity consumption of smart equipment, despite AI-driven automation and intelligence improvements (L. Wang et al. 2024). Therefore, these regions must balance intelligent manufacturing and green innovation with the risk of rebound effects from excessive capital investment. These findings underscore the need for context-specific AI strategies, aligning ET policies with regional structures and factor endowments.

Table 7. Results of moderating effects based on PFS.

	(1) EC	(2) EE	(3) ES
PFS×AI	−0.6083***	4.8607**	−6.1525**
	(0.1086)	(2.4662)	(2.5471)
AI	−0.0009	0.0680***	−0.0503**
	(0.0011)	(0.0209)	(0.0252)
PFS	9.5402***	−126.4762***	38.5942
	(1.9014)	(40.7630)	(36.8599)
Constant	8.6235***	38.6023**	1.4798
	(1.0300)	(17.8444)	(14.3672)
Controls	YES	YES	YES
Id FE	YES	YES	YES
Time FE	YES	YES	YES
N	3654	3654	3654
R^2	0.9263	0.7206	0.6097

Threshold effect

Energy equity reflects equal access, fair distribution, and inclusive participation in the energy transition. This study uses energy equity as a threshold variable to assess AI's threshold effects on energy transition. As shown in Table 8, AI exhibits a double-threshold effect on EC and EE, and a single-threshold effect on ES.

As shown in Table 9, the impact of AI on the energy system evolves through three distinct stages as energy equity (EEQ) improves. In the initial stage ($EEQ \leq v_1$), low EEQ limits AI's effectiveness due to path dependence and market failures. Energy-intensive rebound effects and price distortions lead to increased energy consumption, reduced efficiency, and negligible changes in energy structure. In the intermediate stage ($v_1 < EEQ \leq v_2$), network externalities and synergy effects enable AI to reduce energy consumption, enhance efficiency, and optimize the energy structure by lowering transaction costs and facilitating spillovers. In the advanced stage ($EEQ > v_2$), AI drives energy savings and efficiency through multi-agent coordination, overcoming systemic bottlenecks. Overall, energy consumption and efficiency exhibit greater sensitivity to equity (double thresholds), while structural optimization requires crossing a single institutional barrier (single threshold), suggesting that policy should shift from curbing AI-induced rebound in early stages to enabling structural transformation in later stages.

Table 8. Threshold test results.

			Confidence Interval	
Dependent Variable	v_1	v_2	v_1	v_2
EC	1.6658	5.7727	[1.6516,1.6783]	[5.2609,6.2044]
EE	2.2283	5.7727	[2.2084, 2.2439]	[5.4165,6.0718]
ES	1.6278	—	[1.6038,1.6363]	—

Table 9. Threshold effect regression results.

	(1) EC	(2) EE	(3) ES
$EEQ \le v_1$	0.0076***	−0.0943***	0.0123
	(0.0012)	(0.0210)	(0.0287)
$v_1 < EEQ \le v_2$	−0.0081***	0.5874***	−0.1818***
	(0.0027)	(0.1259)	(0.0463)
$EEQ > v_2$	−0.0729***	3.0966***	—
	(0.0273)	(1.0656)	—
_cons	2.0526***	62.1498***	−22.1205*
	(0.7006)	(15.0364)	(12.5984)
Controls	YES	YES	YES
N	3654	3654	3654
R^2	0.6753	0.2771	0.0580

VI. Further analysis

Model construction

To assess whether the variables are spatially correlated and whether the sample is clustered or dispersed, this study conducts a spatial autocorrelation test. Moran's I, a standard measure of spatial dependence, is used to evaluate the overall spatial correlation of energy transition. The formula is as follows:

$$\text{Index} = \frac{n}{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}} \cdot \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij} z_i z_j}{\sum_{i=1}^{n} z_i^2} \quad (10)$$

where, z_i is the deviation of $(x_i - X)$. $w_{i,j}$ is the spatial weight, which in this study is represented by a distance matrix, with the weight value being the inverse of the distance between the latitudes and longitudes of two regions. n is the total number of factors.

The test results (Table 10) indicate that the energy structure exhibits a pattern of spatial clustering, where high (or low) values are relatively concentrated in neighbouring regions. Moreover, this clustering effect increases year by year. Therefore, spatial factors should be introduced to further study the relationship between AI and ET.

AI, characterized by its permeability, integration, and coordination, can overcome spatial and geographic barriers, thereby generating spatial spillover effects. To examine this, the study employs a Spatial Durbin Model (SDM), which extends the Spatial Autoregressive Model (SAR) and the Spatial Error Model (SEM) by capturing spatial dependencies in both dependent and independent variables. The model is specified as follows:

$$Y_{it} = \rho \sum_{j=1}^{n} W_{ij} Y + \beta_1 AI_{it} + \beta_2 X_{it} + \theta_1 \sum_{j=1}^{n} W_{ij} AI_{it} + \theta_2 \sum_{j=1}^{n} W_{ij} X_{it} + \mu_i + \tau_t + \varepsilon_{it} \quad (11)$$

Where, W_{ij} represents the spatial weight matrix, $\rho \sum_{j=1}^{n} W_{ij} Y$ denotes the spatial spillover effect of energy transition, and $\theta_1 \sum_{j=1}^{n} W_{ij} AI_{it}$ captures the spatial spillover effect of AI. μ_i and τ_t indicate spatial and temporal fixed effects, respectively, while ε_{it} is an independently and identically distributed random error term.

This study uses LM, Wald, and LR tests to select the appropriate model. As shown in the results (Table 11), the model passes the LM test, indicating that both the SEM and SAR models exist, leading to a preference for the SDM model.

Analysis of spatial spillover effects

This study employs the SDM to investigate the spatial spillover effects of AI on the energy transition. The regression results (Table 12) show that while AI significantly promotes the energy transition locally, it inhibits progress in neighbouring regions. This effect is mainly attributable to the early stage of AI development, which leads to a siphoning effect – developed

Table 10. Global Moran's index test.

Year	EC	EE	ES	Year	EC	EE	ES
2006	0.029***	0.006*	0.016***	2013	0.041***	0.008**	0.013***
2007	0.032***	0.008**	0.012***	2014	0.042***	0.013***	0.020***
2008	0.030***	0.004	0.005	2015	0.044***	0.014***	0.026***
2009	0.029***	0.006*	0.011***	2016	0.043***	0.016***	0.032***
2010	0.034***	0.007**	0.014***	2017	0.068***	0.065***	0.018***
2011	0.034***	0.010**	0.019***	2018	0.066***	0.068***	0.013***
2012	0.036***	0.004	0.013***	2019	0.068***	0.123***	0.015***

Table 11. Tests for the SDM model.

	EC	EE	ES
Moran`s I	27.159***	32.677***	27.155***
LM-error	566.488***	823.389***	566.304***
Robust LM-error	485.917***	199.434***	211.858***
LM-lag	106.965***	769.660***	384.002***
Robust LM-lag	26.434***	145.705***	29.556***
LR-sar	65.53***	33.40***	37.77***
LR-sem	57.73***	31.75***	35.65***
Wald-sar	66.02***	33.50***	37.89***
Wald-sem	61.15***	32.34***	35.92***

Table 12. SDM regression results.

	(1) EC	(2) EE	(3) ES
AI	-0.0064***	0.1167***	-0.1194***
	(0.0008)	(0.0173)	(0.0152)
*W*AI*	0.0353***	-0.5043***	0.3846***
	(0.0064)	(0.1470)	(0.1290)
lngdp	0.3934***	0.1272	0.9050
	(0.0418)	(0.9676)	(0.8478)
gov	0.9706***	-12.6117***	10.2481***
	(0.1761)	(4.0726)	(3.5683)
is	0.5980***	-9.4784**	5.5859
	(0.1769)	(4.0912)	(3.5856)
lntra	0.1395***	-2.0691***	1.9126***
	(0.0207)	(0.4793)	(0.4199)
fin	-0.0894***	1.0333	-1.5564***
	(0.0290)	(0.6709)	(0.5879)
rho	0.3075**	0.2778**	0.2727**
	(0.1227)	(0.1257)	(0.1277)
sigma2_e	0.1081***	57.7883***	44.3671***
	(0.0025)	(1.3526)	(1.0381)
N	3654	3654	3654
R^2	0.3852	0.1261	0.0043

regions attract talent, technology, and capital, forming growth poles that enhance local ET but deplete resources in surrounding regions. AI may also drive the relocation of polluting industries to neighbouring regions, worsening their transition challenges. Overall, AI has not yet shifted from siphoning to trickle-down effects, thus failing to benefit nearby regions. This finding contrasts with the results of Song et al. (2024), who find that AI can alleviate neighbouring countries' energy vulnerabilities through spatial spillovers. The divergence stems from the fact that China's application of AI remains primarily efficiency-oriented, lacking scalable models for cross-regional diffusion. These findings underscore several policy challenges: while promoting AI, policymakers must account for its spatial effects by balancing development between core and peripheral regions and avoiding excessive resource concentration.

VII. Conclusion and recommendations

Based on panel data from 261 Chinese cities spanning 2006 to 2019, this study investigates the impact of artificial intelligence (AI) on the energy transition and further explores its mechanisms and spatial spillover effects. The main conclusions are as follows: (1) AI reduces energy consumption, improves energy efficiency, and optimizes the energy structure, promoting energy transition across the dimensions of quantity, quality, and structure. (2) Mechanism analysis confirms that AI enhances human-machine alignment and promotes economic agglomeration, both of which jointly drive the energy transition. (3) Moderation analysis shows that AI is more effective in promoting the energy transition in regions with well-developed digital infrastructure, strong technological absorptive capacity, and labour-intensive factor endowments. (4) Threshold effect analysis reveals an increasing marginal effect, indicating that the impact of AI on the energy transition intensifies as energy equity improves. (5) Spatial spillover analysis reveals significant spatial dependence in the geographical distribution of the energy transition. However, due to the dominance of the siphoning effect over the trickle-down effect, AI currently inhibits the energy transition in neighbouring regions.

Therefore, this study proposes the following policy recommendations. First, promote AI – energy system integration across key domains. For energy consumption, support smart metres and demand-side response systems. For energy efficiency, provide AI-related funding and tax incentives to firms engaged in production optimization. For energy structure, prioritize AI-based grid dispatch in renewable hubs to boost clean energy use. Second, improve human capital and foster economic agglomeration by expanding vocational training and building tech-focused industrial zones for smart production and energy management. Third, enhance infrastructure and strengthen policy support. In low-capacity regions, prioritize infrastructure with central subsidies. In labour-intensive areas, deploy mature energy-saving technologies; in capital-intensive

areas, promote long-term innovations. Fourth, strengthen dynamic equity adjustment tools through tiered fiscal incentives, such as subsidized loans for regions exceeding equity thresholds. Fifth, establish a coordinated regional transition framework. Impose an energy coordination tax on AI-exporting regions to upgrade recipient grid infrastructure and build an AI-driven trading platform to match energy needs and reduce spatial externalities.

This paper also has the following limitations: First, the relatively short sample period, due to data constraints, limits the analysis of AI's long-term effects. Second, the focus on China's prefecture-level cities omits micro- (e.g. firms, households) and macro-level (e.g. cross-country) perspectives. Future research could compare AI-driven energy transitions across China, Europe, the U.S., and Southeast Asia, identify best practices, and assess policy model applicability. Using international data may also help clarify AI's role in global carbon neutrality and its cross-border impacts on energy governance and renewable energy deployment.

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